

Lumen LF-60P Optical Flow Meter Datasheet

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Overview

The Lumen LF-60P is an inline optical-correlation flow meter for clean and lightly loaded liquids in process industry, high-pressure water injection and hot water applications. A pulsed LED light sheet passes through two sapphire windows in the measuring tube, and the electronics cross-correlate the particle and micro-bubble patterns seen by two receivers to calculate flow velocity. With no moving parts and no obstruction in the bore, the LF-60P produces negligible pressure loss and needs no routine recalibration.

The LF-60P is the high-pressure member of the Lumen family. It is rated PN40 (40 bar) and suits pipe sizes from DN25 to DN150. For low-pressure utility work, or for DN200 lines, the LF-60 is usually the more economical choice.

Key features

- Optical correlation measurement with a $\pm 0.75\%$ of reading accuracy across 0.5 to 15 m/s
- Full bore design with no moving parts and no pressure drop worth measuring
- 4-20 mA with HART 7, scaled pulse and Modbus RTU (RS-485) outputs as standard
- 316 stainless steel electronics housing with a 316L measuring tube and sapphire windows
- Integrated process pressure sensor for pressure-compensated density and diagnostics
- Two-line backlit display with local totaliser and diagnostics
- Works in water with turbidity as low as 0.2 NTU

Technical specifications

Parameter | LF-60P value

Nominal sizes | DN25 to DN150

Pressure rating | PN40 (40 bar)

Accuracy | $\pm 0.75\%$ of reading (0.5 to 15 m/s); ± 0.4 cm/s below 0.5 m/s

Velocity range | 0.1 to 15 m/s

Repeatability | $\pm 0.15\%$

Fluid temperature | -20 to +120 °C

Ambient temperature | -40 to +60 °C

Outputs | 4-20 mA with HART 7, pulse (max 2 kHz), Modbus RTU

Power supply | 24 VDC $\pm 10\%$, 5.5 W

Ingress protection | IP67

Housing material | 316 stainless steel

Wetted materials | 316L stainless steel (Hastelloy C-276 option), sapphire windows, FKM seals

Process connection | EN 1092-1 PN40 flanges (ASME Class 300 on request)

Weight (DN50) | 9.8 kg

Straight run required | 15D upstream, 5D downstream

Modbus defaults | Address 1, 19200 baud, 8E1

Current firmware | 1.9.0

Cable entries | 2 x M20 x 1.5

Ordering information

Order codes follow the pattern LF60P-{size}-PN40, where the size is the three-digit nominal diameter. For example, LF60P-050-PN40 is the DN50 meter and LF60P-100-PN40 is the DN100 meter. Add the suffix -HC for Hastelloy C-276 wetted parts (for example LF60P-050-PN40-HC). Prices are published in the current regional price list.

Typical applications

- Boiler feed water and condensate return up to 120 °C
- High-pressure water injection and reverse osmosis feed
- Process water in chemical and pharmaceutical plants
- Flooded valve chambers and outdoor pits where IP67 is required

Installation notes

Install the LF-60P with at least 15 pipe diameters of straight pipe upstream and 5 downstream of the meter. The measuring tube must remain full; on vertical pipes, install with upward flow. Grounding rings are mandatory on non-conductive or lined pipe. Full installation, wiring, HART commissioning and flange bolting instructions are given in the Lumen LF-60P Installation Manual (prod-lf60p-installation-manual).

Compliance

CE and UKCA marked. Pressure Equipment Directive 2014/68/EU, category I for DN25 to DN50, category II for DN65 to DN150. EMC to EN 61326-1 industrial environment. HART registered with FieldComm Group, device revision 3. The LF-60P is not certified for use in hazardous areas.